

Lesson no 3

Need of Professional Ethics in Engineering
(Objectives)

Issues and Sources

Importance of Engineering Ethics as a subject

- Increased awareness of importance due to publicity surrounding high profile engineering failures
- Engineering decisions can impact public health, safety, business practices and politics
- Engineers should be aware of moral implications as they make decisions in the workplace
- Study of ethics help engineers develop a moral autonomy:
 - ability to think critically and independently about moral issues
 - ability to apply this moral thinking to situations that arise in the professional engineering practice

- Ethical problems in engineering are often complex and involve conflicting ethical principles. Engineers must be able to intelligently resolve these conflicts and reach a defensible decision.

Ethical Issues faced by engineers

- Public Safety
- Bribery and Fraud
- Environmental Protection
- Fairness
- Honesty in research and testing
- Conflicts of interest

Five Sources of Professional Ethics

- The Utilitarian Approach
- The Right Approach
- The Fairness or Justice Approach
- The Common good Approach
- The Virtuous Approach

Foundations of Ethics

- Sympathy
- Integrity
- Honesty
- Responsibility
- Charity
- Code of conduct
- Up bringing
- Peace

- Behaviour
- Manners
- Love
- Character
- Truthfulness
- Being honest with your self
- Caring for others
- The Golden rule

Engineers as Managers

- Characteristics

- Promote an ethical climate, through framing organization policies, responsibilities and by personal attitudes and obligations.
- Resolving conflicts by evolving priority, developing mutual understanding and generating various alternative solutions to problems.
- Social responsibility to stakeholders, customers and employers (to produce safe and good products or useful service)

Managing Conflicts

- Conflicts based on schedules
- Conflicts arising out of fixing the priority to different projects or departments
- Conflict based on the availability of personnel
- Conflict over technical, economic and time factors such as cost ,time and performance level
- Conflict arising in administration such as authority, responsibility, accountability and logistics required

- Conflicts of personality, human psychology and ego problems
- Conflict over expenditure and its deviations

Conflict Resolution

- People
- Interests
- Options
- Evaluation

Engineers as advisors in planning and policy making

- Objectivity
- Study all aspects
- Values
- Technical complexity
- National security